GOLDMAN SACHS INTERVIEW EXPERIENCE

About Company and role :

Goldman Sachs is a global investment banking and financial services company, headquartered in New York City. It was founded in 1869 and offers a wide range of services including investment banking, securities, investment management, and other financial services to corporations, financial institutions, governments, and individuals.

Eligible Branches:

UG: AE, AU, AIDS, BM, CE, CS, EE, EC, EI, GI, IE, IT, MN, MS, ME, MI, PT, PE, RP & RA

Eligibility Criteria: 6 & above; No current arrears

Title: Engineering Analyst

Office Location: Bengaluru and Hyderabad

Internship Duration: 8 weeks

Graduation Year: 2027

About Me :

My name is Vijay Anandan J M, and I have maintained a CGPA of 8.56. I began learning Data Structures and Algorithms (DSA) during my first-year summer holidays and started solving basic problems on topics such as arrays and strings on LeetCode from my third semester.

But during my 4th semester I left that habit, since we had full stack development course and were fully concentrating on that and making projects and also we had Design and analysis of algorithms subject which introduced us to advance problem solving techniques like recursion, backtracking, Dynamic programming and Greedy methods which helped me a lot during intern preparation.

Then I started my preparing for intern from mid of our 2nd year summer holidays, just one month before our 3rd year starts.I followed only strivers A-Z dsa sheet and solved problems on my own and at the time of interview I have completed around 75% of that sheet.I used C++ for solving problems. That's all about me.

About Online Assessment and Interview:

Goldman Sachs was the first company that visited our campus and also the first to Keep online assessment.

Online Assessment:

About OA, it was conducted on HackerRank platform. The topics that OA had are, Output guessing from oops both Java and C++ , and about sorting algorithms that is basically theory side of dsa, and had about 3 to 5 Aptitude Questions(just maths problems that we solved during our school days) and one Hr question was also asked, like what you will do in this situation, then finally three coding questions.

Coding Question – 1:

It was based on sorting.

Given an array, e.g., [12, 13, 6], sort it in ascending order according to the number of factors of each element in the array. The output of the question is [13,6,12].

Explanation:

12 → Factors: 1, 2, 3, 4, 6, 12 → 6 factors

13 → Factors: 1, 13 → 2 factors

6 → Factors: 1, 2, 3, 6 → 4 factors

Sorting them by the number of factors (ascending order):

13 → 2 factors

6 → 4 factors

12 → 6 factors

Hence, the output is [13, 6, 12].

Actually this was an very very simple question, our only task was to write optimal algorithm for finding the No. of factors of an element. I wrote that algorithm that finds the factors in O(root N) time where N is the element for which factors are need to be found.

After writing a function for finding the No. of factors, then I used In- build Sort function in C++, and just passes an input array, and an Custom Comparator where the element are compared according to the factors. I used that factors function in this custom comparator.

Hence, within five minutes I was able to solve this problem and all the 15 test cases passed.

Coding Question-2:

I don't exactly remember the question now, It was like N organic substances will be given, each organic substance will have two values like growth factor and additive factor. Our goal was to find the minimum absolute difference Between growth factor and additive factor across all possible combinations of organic substances. How the growth factor and additive factor was calculated for combination of organic substances is the multiplication of all the growth factors in that combination is the new growth factor for that whole combination and similarly the addition of the additive factors in that combination is the new additive factor for that combination.

I solved that problem using Recursion first, like just finding all the possible subsequences

While finding itself will multiply the growth factor and add additive factor and then finding the minimum once end index is reached. But that solution was not accepted for some test cases. Then later I solved this using power set algorithm which is used for finding subsequences. Both take same time c but space c differs since power set doesn't use recursion stack space just uses bit manipulation and loops.

Then for this problem also all the 15 test cases was passed after using that power set algo.

Actually I didn't read this question, I directly went to sample input and output and I was able to understand what we have to do and find.

Anyone who solved subset sum problem can solve this problem to.

Coding Question – 3:

In this question , we will be given with new employees and their work experience and skill score as string [][3] format like,

E 1 2

like this will be given in string format, [x][0] will be either E or Q, E stands for employee.

And [x][1] ,[x][2] represents score and work exp respectively for that employee. These employees can be assigned ids from 1 and goes on just have to assume like that. After that, Q stands for query, it will be also present in that string[][3] itself, like

E 1 2

E 2 3

E 3 4

Q 2

E 5 6

Q 3

This Q 2 means that the employee 2 that E 2 3 needs a mentor for him and our task is to find the employee among the employees given using the criteria like, skill and exp should be greater than the employee who needs a mentor. If multiple employees satisfy the condition, return the employee who has least diff of skill score and work exp between the employee who needs a mentor and the employee who can be assigned as a mentor.

Was asked to return the ans in vector of strings.

I coded this problem for nearly 20 min and was able to pass only 7 test cases.

Then I just submitted and started solving MCQs and aptitude questions.

After few the days, the results were announced, around 28 students were selected for interview and I was one among them.

Interview Round-1 :

This is my first interview, and it was held at CUIC, and I was also very nervous. But the interviewer spoke well and was good.

First, he asked me to give a brief introduction about myself. Then he started with a basic aptitude math question, like when two dice that have 6 faces are rolled, what is the probability of the sum of the faces being prime? I wrote the sample space on the paper, I was explaining the answer, and he was satisfied with that answer. Then the follow-up question was, if we roll dice with N faces for M times, what will be the probability of the sum of faces after the dice are rolled for M times being prime? I was asked to solve it in a programmatic way and was also asked to write the pseudocode. I wrote the code and also explained my approach. I just solved that problem using recursion, that is brute force, just trying out all the possible ways, and while finding, I would just add the sum, and after we reach M+1 times, I would stop and check whether the sum is prime or not. And after counting, I would just divide that count by all the possible combinations, which is N^M, to find the probability. I just explained this and wrote the pseudo code.

He was satisfied with my answer. And in the follow-up question, he asked the time and space complexity of the code I wrote and I also said that, and he was satisfied with those answers too.

Then he asked me whether I had any questions for him. I just asked about the tech stack they use, since many of our seniors also asked this same question (I read it in the placement experiences PDF). That’s all about round 1. Overall It went well.

Out of 28 people, nearly 18 were eliminated in this round itself.

Interview Round -2:

Immediately after I had lunch around 1 o clock, I was called for Round 2. Since I had already attended Round 1, I was a bit more confident.

First, he asked me to give a brief introduction about myself, and then he directly started diving into my projects. He took one of my projects and asked me to explain what we did, etc. He asked me to write the tables that we created in the database for that project and to explain the difficulties we faced, as well as how I normalized that table. I explained everything.
After that, from one of the tables I explained, I was asked to write a query. That query was of medium difficulty, involving GROUP BY, ORDER BY, and selecting the first 10 rows. I did that, and then again, a simple programming question based on the sliding window concept was given. I was asked to write the pseudocode and also explain the time and space complexities. I did that too, and overall, he was satisfied. And I was sure that will get selected for round 3 and also started preparing for it.

Interview Round – 3:

After round 2, within 10minutes I was called for next round. In this round, one mam was there, actually this was also an technical round , I thought It would be an HR round. I was asked to explain only about the projects I did. She asked all the possible questions that can be asked from my projects. I actually prepared for those questions before the interview(like I asked LLMs like what are all the possible questions that can be asked for these projects, and also asked that itself to give answers) I just read that once and Luckly she also asked only those questions that LLM's said, I answered for every questions. This round went round 45 minutes and At last she asked me whether I have any questions and I asked what is the recent challenge that your team faced and how you solved them. She was answering to my question, which I didn't listen. And then just said thank you and left. My round 3 interview was completed at around 2.30 and I was asked to wait.

I was asked to wait till 6.15 and after 6.15, the students who were asked to wait were called by the HR and the team and they just said that we are selected and congratulated us.

After round 2, only 3 people was called for round 3 and those three were selected.

Out of 28 students, 3 were selected and I was one among them, and remaining two were from MIT campus.

Preparation tips:

- I referred Abdul Bari on Udemy for theory concepts of DSA.
- I completely followed only the strivers A-Z dsa sheet for problem solving and practiced on Leetcode and in GFG and also participated in C.O.D.E series contest to test my skills. Sheet link: https://takeuforward.org/strivers-a2z-dsa-course/strivers-a2z-dsa-course-sheet-2
- Do projects on your own during your 3rd and 4th semesters , even if you didn't, just learn the project well that's enough to crack interview.
- Try to solve problems on your own, just don't directly see the solution, even if it takes more time just try until you can.
- Participate in C.O.D.E series by ACM, attend the meeting conducted by them, listen to their advices.
- Make use of summer holidays given after 2nd yr.
- In that summer holidays, itself spend atleast one to two hours for aptitude too. Anyone can solve aptitude questions but the one who solves first matters, hence practice that too. I spent 2 hrs for aptitude and 4 to 6 hrs for coding daily.
- Don't lose hope at any moment. Believe in persistency.

Hope anyone who is reading this, Will write like me one day. All the best, Guys.

For any doubts, feel free to contact,

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